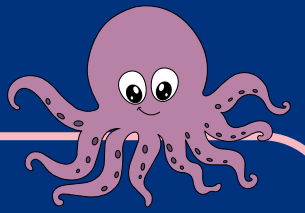


EIGHT ARMS, MANY MINDS: HOW OCTOPUS ARMS THINK FOR THEMSELVES



OVERVIEW

Octopus arms are controlled by a segmented nervous system where most sensory processing and motor control occur locally in the arms and not the central brain

1

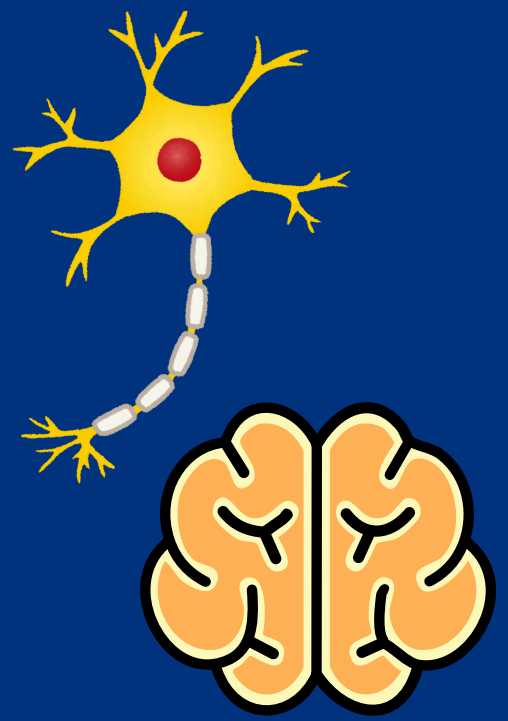
Nervous System Structure

Neuron distribution:

- Total neurons: approximately 500 million
- It has 9 brains but only 10% of neurons are located there!
- Over 60% are located in the arms!

Each arm contains:

- An axial nerve cord
- Peripheral ganglia
- Networks linking hundreds of suckers

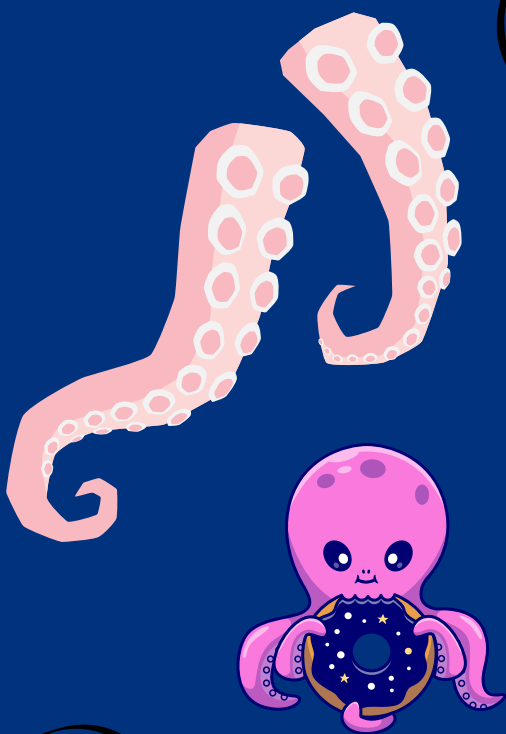


2

Sensory Processing

Each sucker contains:

- Mechanoreceptors for touch and pressure
- Chemoreceptors for taste and chemical detection
- Information is processed locally in the arm before reaching the brain
- As a result, the arms can identify things such as: whether something is edible or its texture



3

Motor Control

Octopus arms:

- Have no bones and have infinite degrees of freedom
- Can bend, twist, elongate and contract independently
- IMAGINE: should an octopus have centralised control, continuous position tracking and massive processing would be required

Think.... 🧠💻 Trying to control an octopus from the brain is like trying to manually animate every pixel in a video game



4

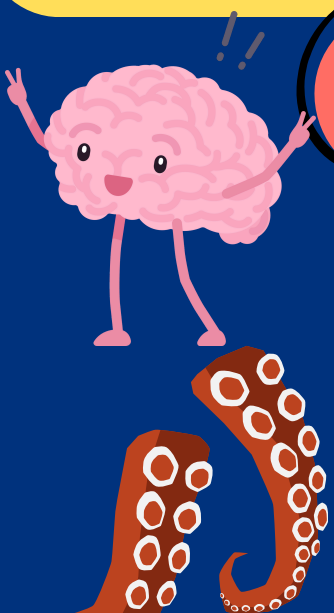
Decentralised Control in Action

What the brain does:

- Sets the intention (e.g. find food)
- Coordinates large scale behaviour

What the arm does:

- Explores environment
- Chooses movements
- Adjusts grip in real time



Call to action: Don't be so single minded. Even octopus aren't!